

```
In [1]: import numpy as np
import pandas as pd
import matplotlib as plt

x = [5, 9, 2, 3, 4, 6, 7, 0, 8, 12, 2, 9]

print(x[1])

print(x[1:4])

print([x[i] for i in [1,2,5]])

print([x[i] for i in range(0,5)] + [x[i] for i in range(9,12)])

print([x[i] for i in range(len(x)) if i not in range(10,12)])
```

```
9
[9, 2, 3]
[9, 2, 6]
[5, 9, 2, 3, 4, 12, 2, 9]
[5, 9, 2, 3, 4, 6, 7, 0, 8, 12]
```

```
In [ ]: #2. Milk Sales - Statistical analysis
y = [33,44,29,16,25,45,33,19,54,22,21,49,11,24,56]

data = np.array(y).reshape(5,3)

df = pd.DataFrame(data, columns=["Shop1", "Shop2","Shop3"], index=["Mon", "Tue","Wed", "Thu", "Fri"])

print(df)

print("\n Statistical Summary for Each Day \n")
print(df.apply(lambda r: r.describe(), axis=0))

print("\n Statistical Summary for Each Shop \n")
print(df.apply(lambda r: r.describe(), axis=1))
```

	Shop1	Shop2	Shop3
Mon	33	44	29
Tue	16	25	45
Wed	33	19	54
Thu	22	21	49
Fri	11	24	56

Statistical Summary for Each Day

	Shop1	Shop2	Shop3
count	5.000000	5.000000	5.000000
mean	23.000000	26.600000	46.600000
std	9.924717	10.014989	10.737784
min	11.000000	19.000000	29.000000
25%	16.000000	21.000000	45.000000
50%	22.000000	24.000000	49.000000
75%	33.000000	25.000000	54.000000
max	33.000000	44.000000	56.000000

Statistical Summary for Each Shop

	count	mean	std	min	25%	50%	75%	max
Mon	3.0	35.333333	7.767453	29.0	31.0	33.0	38.5	44.0
Tue	3.0	28.666667	14.843629	16.0	20.5	25.0	35.0	45.0
Wed	3.0	35.333333	17.616280	19.0	26.0	33.0	43.5	54.0
Thu	3.0	30.666667	15.885003	21.0	21.5	22.0	35.5	49.0
Fri	3.0	30.333333	23.158872	11.0	17.5	24.0	40.0	56.0

```
In [13]: # 3. Matrix Operations
x = np.array([[3,2], [-1,1]])
y = np.array([[1,4,0], [0,1,-1]])

print(" 2* x \n", 2* x)
print("\n x* x \n", x* x)
print("\n x %*% x \n", x @ x)
print("\n x%*% y \n", x @ y)
print("\n Tanspose \n", y.T)
print("\n Inverser \n", np.linalg.inv(x))
```

```
2* x
[[ 6  4]
 [-2  2]]
```

```
x* x
[[9 4]
 [1 1]]
```

```
x %*% x
[[ 7  8]
 [-4 -1]]
```

```
x%*% y
[[ 3 14 -2]
 [-1 -3 -1]]
```

```
Tanspose
[[ 1  0]
 [ 4  1]
 [ 0 -1]]
```

```
Inverser
[[ 0.2 -0.4]
 [ 0.2  0.6]]
```

In [15]: *# 4. Subscript Operations*

```
x = np.array([[3,2], [-1,1]])
y = np.array([[1,4,0], [0,1,-1]])

print("\n a) x[1,] --> Print First Row \n", x[0,:])
print("\n b) x[2,] --> Print Second Row \n", x[1,:])
print("\n c) x[,2] --> Print Second Column \n", x[:,1])
print("\n d) y[1,2] --> Element at row1 col2 \n", y[0,1])
print("\n e) y[,2:3] --> Print Column 2 to 3 \n", y[:, 1:3])
```

a) `x[1,]` --> Print First Row
`[3 2]`

b) `x[2,]` --> Print Second Row
`[-1 1]`

c) `x[:,2]` --> Print Second Column
`[2 1]`

d) `y[1,2]` --> Element at row1 col2
`4`

e) `y[:,2:3]` --> Print Column 2 to 3
`[[4 0]
[1 -1]]`